

CONSTRUCTION MODULE

Construction Inventory Management System

A complete end-to-end inventory control, procurement tracking, and material accountability solution for construction companies managing multiple active sites — going far beyond what traditional systems offer.

-  Multi-Site Stock Control
-  Full Procurement Cycle
-  Real-Time Analytics
-  Theft Prevention
-  Finance Integration
-  Offline Mobile App

8

Core System Roles

6

Procurement Stages

∞

Sites Supported

30+

Intelligence Features

The Problem We Are Solving

SECTION 01

WHY THIS SYSTEM EXISTS

The Triple Inventory Nightmare

Every construction company faces the same three unresolved questions about their materials every single day.



Where is the material?

Is it at Site A, Site B, the central warehouse, in transit, or quietly disappeared? Nobody knows for certain.



Was it actually used?

You bought 200 bags of cement. Only 160 are accounted for. The foreman has no answer. Money has walked out the gate.



Did the price make sense?

Is your procurement officer buying at market rate or inflating figures? Without a benchmark, nobody knows until it is too late.



Industry Reality Check

Research shows uncontrolled material waste and theft account for **15–25% of total project costs** in unmonitored operations. For a ~~£~~**£500 million** project, that is up to **£125 million lost** — purely from poor inventory management. This system closes that gap permanently.

WHAT THIS SYSTEM DOES

A Complete Material Accountability Engine

From the moment a purchase is requested to the moment the last bag of cement is used on-site, every gram of material is tracked, validated, and reported on — across every site you operate.

100%

Material traceability from PO to site use

3-Way

PO to GRN to Invoice matching before payment

Live

Stock levels across all sites simultaneously

AI-Flag

Automatic anomaly and theft pattern detection



Built for Multi-Site Construction Operations

Whether you have 2 active sites or 20, this system handles them all from one unified dashboard. Each site operates independently with its own store keeper, stocks, and reporting — while management has a complete bird's-eye view across the entire portfolio at all times.

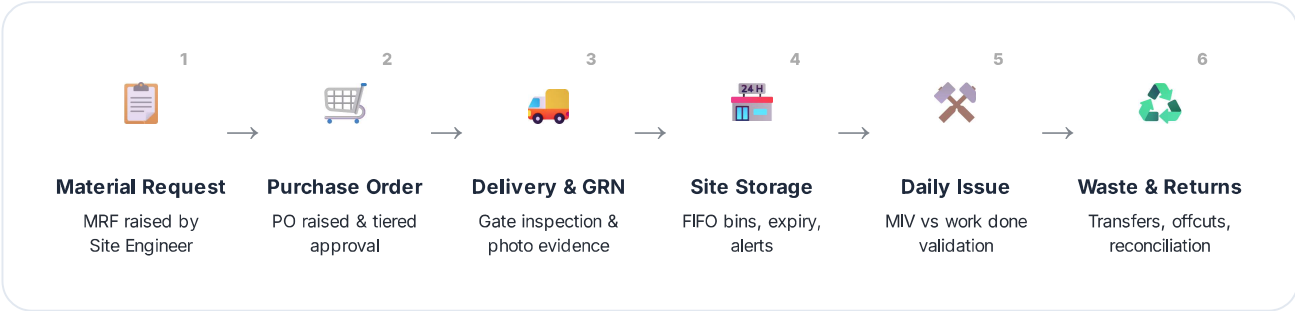
The Full Procurement Lifecycle

SECTION 02

END-TO-END MATERIAL JOURNEY

From Request to Site Use — Nothing Skipped

Most systems track stock levels. This system tracks the entire journey of every material across 6 mandatory stages — with approvals, validations, and evidence at each step.



STAGE 1 — MATERIAL REQUISITION FORM (MRF)

- ✓ Site Engineer selects the specific work activity (block work, concrete pour, roofing)
- ✓ System auto-suggests quantity based on Bill of Materials vs current stock on that site
- ✓ Budget check runs automatically — flags if request exceeds approved activity budget
- ✓ Project Manager receives instant notification for approval or rejection with notes

STAGE 2 — PURCHASE ORDER (PO) GENERATION

- ✓ Approved MRF auto-converts to a draft Purchase Order with supplier selection
- ✓ Last 3 purchase prices for this material shown alongside current market price index
- ✓ Multiple supplier quote comparison before PO is finalized
- ✓ PO validity window — auto-expires if supplier fails to deliver within agreed days

PO APPROVAL TIERS — VALUE-BASED AUTHORIZATION

TIER 1	₦0 — ₦500,000	Project Manager can approve directly from their phone
TIER 2	₦500,001 — ₦5,000,000	Managing Director email/app notification with approve button

TIER 3

Above ₦5,000,000

 **Board / Executive Committee** formal approval required with minutes reference

Delivery Inspection & On-Site Storage

SECTION 03

STAGE 3 — GOODS RECEIVED NOTE (GRN)

Every Delivery is an Evidence Event

A delivery truck arriving at your site is one of the highest-risk moments in your operation. The GRN process turns every delivery into a tamper-proof digital evidence trail.

**QR Scan at Gate**

- Driver presents delivery note QR code at gate
- Store Keeper scans with mobile app instantly
- System auto-loads the matching PO for verification
- Unknown deliveries are flagged immediately

**GPS-Tagged Photo Evidence**

- Photos taken of every delivery — mandatory requirement
- GPS coordinates + timestamp embedded automatically
- Delivery must occur within GPS geofence of the site
- Photo uploads from outside site boundary are rejected

**Quantity Variance Detection**

- PO says 500 bags — GRN records actual received count
- Any variance auto-flags payment and raises a dispute
- Rejected/damaged items tracked on return-to-supplier log
- Supplier gets debit note for short deliveries automatically

**Batch & Lot Tracking**

- Batch/lot numbers recorded for all traceable materials
- Cement: manufacture date + batch number entered at delivery
- Enables FIFO enforcement on issue automatically
- Quality failure traced back to exact batch if issue arises

STAGE 4 — ON-SITE STORAGE MANAGEMENT

Your Site Store is Now a Smart Warehouse

**Company HQ / Central Warehouse — Master Stock Point**

**Site A — Lekki Phase 1 Housing (Project: LPH-001)**

Main Site Store (locked, store keeper managed)

Active Use Zone (today's poured concrete, in-progress work)

Waste / Offcut Pile (tracked separately: avoidable vs unavoidable)

 **Site B — Abuja Commercial Complex (Project: ACC-002)**

 Ground Floor Store (primary store)

 Floor 3 Sub-Store (materials pre-positioned for upper floors)

 **Site C — Port Harcourt Access Road (Project: PH-RD-003)**

 Open Yard Storage Zone A (aggregates, laterite, sand)

 Secure Container Store (high-value materials)

SMART STOCK ALERTS

 **Critical Alert:**

Stock below safety threshold — purchase required immediately

 **Reorder Alert:**

Approaching minimum level — initiate MRF within 48 hours

 **Expiry Alert:**

Cement batches older than 90 days flagged for quality test

INTER-SITE TRANSFER ENGINE

 Site A has excess iron rods; Site C needs them — transfer raised digitally with dual confirmation

 Auto-generates matching Issue Voucher (Site A) + Received Note (Site C)

 Cost allocation moves with the material — finance records update automatically

Daily Material Issue, Waste Tracking & Reconciliation

SECTION 04

STAGE 5 — MATERIAL ISSUE VOUCHER (MIV)

Every Bag Issued is Linked to a Work Activity

Material issue is where most systems have a complete blind spot. When a foreman collects cement from the store, the system must know exactly what that cement was used for — and verify the amount makes sense.

- 1

Foreman Requests Issue

Opens app, selects today's work activity e.g. "Column concreting, Grid A-B"
- 2

Quantity & BOM Check

System checks: does requested quantity match standard consumption for the stated volume of work?
- 3

Store Keeper Confirms Issue

Physical handover confirmed in app — both parties sign digitally on device
- 4

Real-Time Balance Update

Stock balance deducted instantly across system — available to all stakeholders live
- 5

End-of-Day Reconciliation

Physical count vs system count. Any gap triggers mandatory explanation from store keeper



Consumption Rate Violation Example

Standard rate: **6 bags cement per m³ of concrete**

Bricklayer requested: **12 bags for 1m³**

System flags: **"Consumption 100% above standard rate."** Project Manager is notified. Foreman must provide written explanation before issue is approved.



Bill of Materials (BOM) Engine

The system holds standard consumption rates per material per unit of work — set by your QS. These are the baseline against which all daily issues are validated automatically, eliminating guesswork and preventing over-issue.

STAGE 6 — WASTE, RETURNS & LOSS CLASSIFICATION

Every Gram of Waste is Categorized



Avoidable Waste (Preventable)

- Cement spilled through careless handling
- Tiles cracked due to poor stacking
- Materials left in rain or weather without cover
- Over-mixing or wrong mix ratios by labour
- Assigned to responsible foreman team for accountability



Unavoidable Waste (Acceptable)

- Tile offcuts from fitting around corners
- Reinforcement bar trimmings at joints
- Mortar residue after block-work sessions
- Standard shrinkage during concrete curing
- System compares against industry tolerance benchmarks



Material Returns



Loss Reporting

- Excess materials returned to store same day
- Reason for return must be logged by foreman
- Returned materials go back into stock immediately
- Frequent returns may indicate poor activity planning

- Weather damage and site accidents documented with photos
- Insurance-compatible loss report auto-generated
- Finance notified immediately for budget cost adjustment
- Full audit trail maintained for every loss event

Intelligence, Analytics & Anomaly Detection

SECTION 05

THE INTELLIGENCE LAYER

The System Thinks So You Don't Have To

Beyond tracking stock levels, this system continuously analyzes patterns, compares against benchmarks, and proactively surfaces issues before they become expensive problems.



Variance Analysis Engine

Compares theoretical consumption (BOM-based) vs actual consumption per site, per week, per foreman team. Highlights who and what is over-consuming and by how much.



Supplier Performance Scorecard

On-time delivery rate, short-delivery frequency, price consistency vs invoice, and quality rejection rate — all automatically scored and ranked per supplier.



Cash Flow Projection

Based on your project schedule and BOM, forecasts material spend for next 30, 60, and 90 days. Finance team is never surprised by a large purchase requirement again.



Waste Heatmap

Which site wastes the most? Which material? Which foreman team? Visual heatmap shows waste concentration — enabling targeted supervision and training interventions.



Price Index Benchmarking

Compares your purchase prices against Lagos, Abuja, and Port Harcourt market price indices monthly. Flags when you are overpaying by more than 10% vs market rate.



Profitability Impact Tracker

Material cost as % of project budget — live. Cost per m² of completed work by section. Feeds directly into project P&L so accounts are always current and accurate.

FRAUD & THEFT PREVENTION

The System Catches What Eyes Miss



Ghost Delivery Detection

PO raised, GRN raised, payment made — but no vehicle GPS trace ever entered the site. Flagged immediately.



Perfect Match Suspicion

Real deliveries almost always have minor variances. If GRN always matches PO exactly — that pattern itself is flagged as suspicious.



After-Hours Delivery Flag

High-value materials delivered outside working hours are automatically escalated to the Project Manager and MD for review.



Staff Pairing Pattern

If the same driver always delivers when a specific store keeper is on duty, the system flags this recurring pattern for investigation.



Progress vs Issue Gap

Materials issued but no matching work progress logged. Cement was "used" but no concrete was poured according to the site diary.

**Friday Afternoon Anomaly — A Real Example**

The system detected that waste rates at Site C spiked every Friday afternoon between 3PM–5PM. Investigation revealed that materials were being bundled into "waste" logs to cover unauthorized removal before the weekend. This pattern is invisible to human supervisors but obvious to the system's weekly trend analysis.

System Roles, Access Control & Finance Integration

SECTION 06

WHO USES THIS SYSTEM

Role-Based Access — Right Information, Right Person

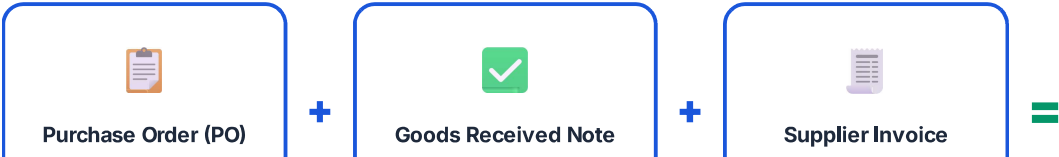
Every person in the system has a defined role with specific permissions. Nobody sees what they should not see. Everyone has exactly what they need to do their job well.

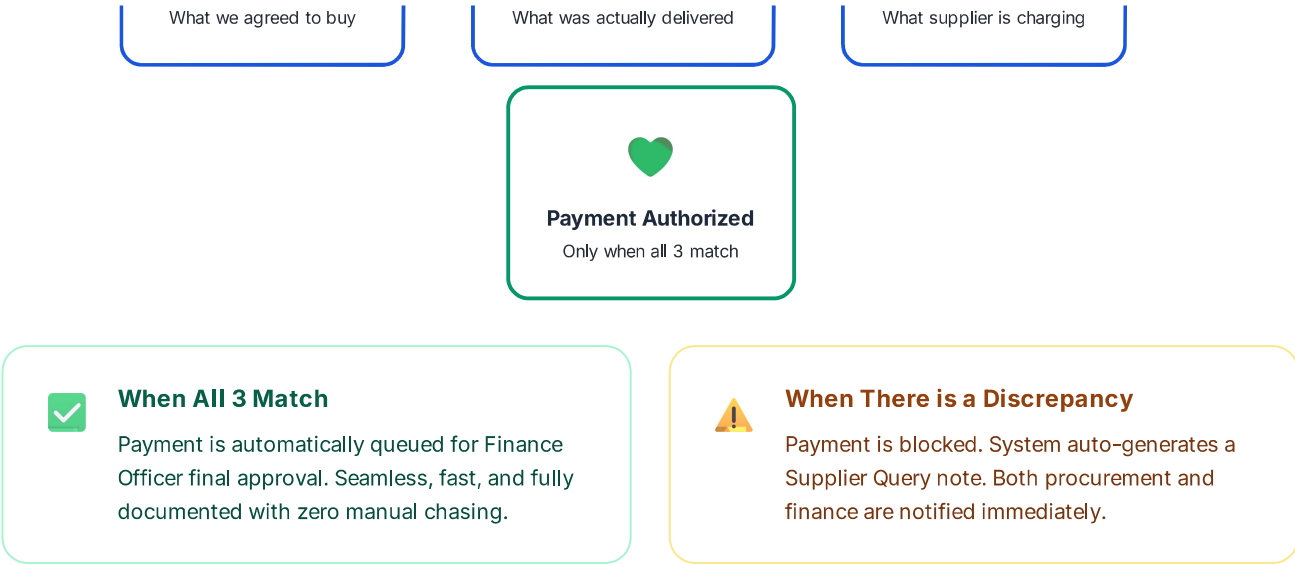
Role	Primary Responsibility	Key System Actions	Level
MD / Director	Executive oversight & high-value approvals	Approve POs > ₦500k, view anomaly alerts, cross-project P&L dashboard	Executive
Procurement Officer	PO management & supplier relationships	Create POs, manage supplier database, run price comparisons, track deliveries	Procurement
Project Manager	Site-level operations approval	Approve MRFs, approve POs to ₦500k, view site dashboards and variance reports	Management
Quantity Surveyor	Standards, BOMs and financial validation	Set/update BOM consumption rates, run variance audits, validate invoices against QS rates	Technical
Site Engineer	Field operations and material requests	Raise MRFs, log daily work progress, approve MIVs, submit site diary entries	Operational
Site Store Keeper	Physical stock custody & documentation	Receive goods (GRN), issue materials (MIV), daily count reconciliation reports	Operational
Finance Officer	3-way match & payment authorization	Match PO to GRN to Invoice, approve payments, view cost reports and budget tracking	Finance
Auditor	Compliance & trail verification	Read-only access to complete audit trail — cannot create or edit any record	Read-Only

FINANCE INTEGRATION — THE 3-WAY MATCH

No Match = No Payment. It Is That Simple.

Before a single naira is paid to any supplier, three documents must align perfectly: the original Purchase Order, the Goods Received Note, and the Supplier Invoice.





Mobile App, Integrations & Hidden Power Features

SECTION 07

FIELD-FIRST MOBILE DESIGN

Built for Construction Sites, Not Office Desks

Most software assumes strong internet, a laptop, and a trained IT department. Construction sites have none of these. This system was designed for how sites actually operate — not how they should operate in theory.



Offline-First Architecture

The mobile app works completely offline. Store Keepers log GRNs, issue materials, and do reconciliations even without network. Data syncs automatically when connection returns.



Budget Android Device Support

Designed to run smoothly on low-cost Android devices in the ₦30,000–₦50,000 range. No iOS dependency. No expensive device procurement required.



Voice-to-Text Input

Store Keepers who struggle with typing can dictate entries. The system transcribes and structures the data automatically. Removes literacy as a barrier to adoption.



WhatsApp-Style Approvals

Approval notifications arrive like familiar WhatsApp messages. Project Managers can approve MRFs with a single tap from their phone — no full login required for basic approvals.



QR Label Printing

Print QR code labels for material bags, pallets, and storage bins directly from the system. Enables instant scanning and identification of any item at any time.



Rainy Season Mode

When activated, automatically adjusts expected waste tolerance rates for weather-sensitive materials. Prevents false waste alerts during the wet season.

MODULE INTEGRATIONS

Connected Across Your Entire Business



Project Schedule (Gantt)

Material need dates linked directly to activity start dates. System pre-orders materials before they are needed based on your project programme.



Finance & Accounts

POs auto-generate a liability entry. GRNs trigger the payment queue. 3-way match required before invoice is processed. Zero double-entry.



HR & Labour Management

Foreman teams linked to material issues. Labour accountability — if a team signs for materials, they own accountability for proper use of those materials.



Quality Control

Material batch numbers linked to test results. Pass/fail quality certifications attached to stock records. Failed batches blocked from issue automatically.



Equipment & Plant



Supplier Portal

Diesel, engine oil, welding rods, generator fuel — all tracked with the same system. Equipment consumables managed alongside construction materials.

Suppliers log in to see their own POs, submit invoices digitally, track payment status, and view dispute notices. Reduces back-and-forth calls significantly.



Subcontractor Accountability

Subcontractors sign digitally for materials handed to them. They accept accountability for proper use. Any loss becomes a contractual charge-back item.



Client Transparency Portal

Filtered view showing clients exactly what materials were bought for their project — builds exceptional trust and is a powerful competitive differentiator.

Implementation Roadmap & Getting Started

SECTION 08

ROLLOUT STRATEGY

A Phased Approach for Zero Disruption

The system goes live in phases — not a big-bang switch-over that disrupts active projects. Each phase builds on the last, and teams are trained incrementally with support at every step.

01

Weeks 1–2

Foundation Setup

Configure master material catalogue, upload Bills of Materials for existing projects, create site structures, register users and assign roles. Import existing stock counts as opening balances.

02

Weeks 3–4

Pilot Site Launch

Roll out to one active site first. Train Site Engineer, Store Keeper, and Project Manager. Run all 6 procurement stages in parallel with the current paper process for 2 weeks to validate accuracy before cutting over.

03

Weeks 5–6

Full Procurement Go-Live

Cut over to fully digital procurement on pilot site. All POs, GRNs and MIVs are now system-only. Finance team begins 3-way match process. Paper process retired for this site permanently.

04

Weeks 7–10

All-Sites Rollout & Intelligence Activation

Expand to all active sites. Activate intelligence modules: variance analysis, supplier scoring, cash flow projections, and anomaly detection. First full cross-site management report generated.

WHAT YOU NEED TO PREPARE

Your Readiness Checklist

 List of all active construction sites with project codes and site boundaries

 Standard material list for your typical project types (from QS or past projects)

 Bill of Materials or Quantity Surveyor to help define standard consumption rates

 Android phones for each Site Store Keeper (minimum one per site)

 Current approved supplier list for loading into the supplier database

 Current open POs and stock balances per site for opening balance entry

 List of staff who will use the system, with their intended roles assigned

 Finance approval tiers and authority limits confirmed in writing by MD



Ready to Build This Into NAWPropertyFlow CRM?

This module integrates directly into your existing NAWPropertyFlow CRM platform — leveraging your existing Properties, Projects, Finance, and HR modules. No separate system to manage. One login. One platform. Complete construction operations visibility from day one.

CONSTRUCTION INVENTORY MANAGEMENT SYSTEM

Every Material. Every Site. Every Naira. Accounted For.

Stop losing 15–25% of your project costs to untracked waste, ghost deliveries, and procurement fraud. The system pays for itself before the first project is complete.

6

Procurement Stages

8

System Roles

∞

Sites Supported

30+

Intelligence Features

NAWPropertyFlow CRM

Enterprise Real Estate & Construction Management Platform

 demo.nawpropertyflow.com.ng info@nawpropertyflow.com.ng Construction Module Enquiries

